VV-S-190
October 26, 1961
SUPERSEDING
(See 6.3)

FEDERAL SPECIFICATION

SEALING COMPOUND (DIPCOAT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one grade of sealing compound suitable for the sealing of wrappings or packages by dipcoating.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

NN-K-231—Kegs, Wood, Slack.

QQ-A-355—Aluminum Alloy, Plate and Sheet 2024.

QQ-C-576—Copper Plates, Rolled Bars, Sheets, and Strips.

QQ-S-636—Steel; Carbon (Low-Carbon), Sheets and Strips.

PPP-B-41—Barrels; Wood, Slack.

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box. Fiberboard.

PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Test Method Std. No. 101—Preservation, Packaging, and Packing Materials: Test Procedures.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-B-121—Barrier Material, Greaseproofed, Flexible (Waterproofed).

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials:

ASTM Method D5—Test for Penetration of Bituminous Materials.

ASTM Method D127—Test for Melting Point of Petrolatum and Microcrystalline Wax.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Material. The material covered by this specification shall be homogeneous, stable, and noncorrosive. The raw materials shall be such as to produce a compound meeting the requirements of this specification. The manufacturer is given wide latitude in the selection of raw materials and process of manufacture. The finished materials when either in a liquid or solid state shall be free from all foreign substances. The material shall not be toxic to personnel under normal conditions of usage (see 6.4).

3.2 Physical requirements.

3.2.1 *Softening point.* The minimum softening point of the compound shall be 150°F. when tested as specified in table I.

3.2.2 *Penetration.* The maximum penetration shall be 40 at 77°F. (25°C.) when tested as specified in table I.

3.2.3 *Cohesion at low temperatures.* A coating of the compound shall show no cracking at a temperature of minus 10°F., nor shall it show discontinuities when cooling from the liquid to the solid state when tested as specified in table I and 4.5.1.

3.2.4 *Contraction on solidification.* The compound shall show no checking, cracking, or retraction that would permit entry of water into the package when tested as specified in table I.

3.2.5 *Homogeneity.* The compound shall show no separation and there shall be no residue when tested as specified in 4.5.2.

3.2.6 *Water absorption.* The compound shall not increase more than one percent by weight when tested as specified in 4.5.3.

3.2.7 *Corrosion.* The compound shall be noncorrosive to steel, copper, and aluminum when tested as specified in 4.5.4.

3.2.8 *Setting time.* A $\frac{1}{64}$-inch coating of the compound shall set within one and one-half (1.5) minutes when testesd as specified in 4.5.5.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. For the purpose of sampling and inspection a lot shall consist of one factory batch of not more than 10,000 pounds when inspection is made at the place of manufacture. When inspection is made at the place of delivery a lot shall include all material in a single shipment.

4.3 Sampling.

4.3.1 *Sampling for inspection of filled containers (at the contractor's plant).* A random sample of filled containers shall be selected in accordance with Military Standard MIL-STD-105 with acceptable quality level (AQL) equal to 2.5 percent defective to verify compliance

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with all stipulations of this specification in regard to fill, closure, and other requirements not involving tests.

4.3.1.1 When Military Standard MIL–STD–105 specifies an action by the Government, it shall at the option of the Government be performed either by the Government or by the contractor under the surveillance of the Government inspector.

4.3.2 *Sampling for tests.* When inspection is made at the place of manufacture, a ten-pound sample shall be taken from each lot. When the material is inspected at the place of delivery, ten percent but not more than ten nor less than five of the containers in a lot shall be selected for sampling. An approximately ⅛-pound sample shall be taken from the upper, middle and lower portion of each container selected for sampling. If the lot is contained in fewer than five containers, each container shall be sampled. The sample from each container shall weigh approximately one pound. The one-pound samples so selected shall be placed in a suitable container and heated until the wax is liquid, but at no time above a temperature of 230°F. Agitate the wax until a homogeneous mixture is obtained. Cast the wax into a mold. One half shall be forwarded to the laboratory specified by the procuring service. The remainder shall be retained in a clean dry sealed container by the inspector for future reference.

4.4 Inspection.

4.4.1 *Inspection of filled containers (at the contractor's plant).* Unless otherwise specified, inspection of filled containers shall be in accordance with Military Standard MIL–STD–105. Each sample filled container as specified in 4.3.1 shall be examined for defects of construction of the container and the closure. Each sample filled container shall also be checked to determine the amount of contents. Samples shall be inspected for compliance with the packaging, packing and marking requirements of section 5.

4.4.2 *Lot acceptance tests.* The samples selected in accordance with 4.3 shall be subjected to the tests specified in 4.5.

4.5 Test procedures. The tests shall be conducted in accordance with table I and as hereinafter specified.

TABLE I.—*Index of tests*

Requirement	Test method		Section of this specification giving requirements
	Test method of ASTM	Section of this specification giving further reference	
Softening point	ASTM D127	....	3.2.1
Penetration	ASTM D5	....	3.2.2
	Test method of Fed Std. No. 101		
Cohesion at low temperature	193	4.5.1	3.2.3
Contraction on solidification	103[1]	....	3.2.4
Homogeneity	...	4.5.2	3.2.5
Water absorption	...	4.5.3	3.2.6
Corrosion	...	4.5.4	3.2.7
Setting time	...	4.5.5	3.2.8

[1] Barrier material used for wrapping the boxes shall conform to type 1, grade c of Military Specification MIL–B–121.

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4.5.1 *Cohesion at low temperature.* The paper used in this test shall be standard 50-pound kraft paper. After testing the sealing compound in accordance with method 193 of Fed. Test Method Std. No. 101, the compound shall be examined for cracking. Any cracking on more than one strip shall be cause for rejection. Cracking is defined as any fracture or break in the coating.

4.5.2 *Homogeneity.* An approximate 200 g. sample of the sealing compound shall be placed in a tared 400-ml. Pyrex beaker and heated in a convection current type oven at 210° ± 5° F. for 72 hours. The beaker shall then be removed from the oven and visually examined while hot for separation into layers of different appearance or separation of solid, semisolid or liquid matter. The molten material shall be poured out and the beaker examined for evidence of residue.

4.5.3 *Water absorption.* Three empty chipboard boxes shall be wrapped and dipcoated as specified in the test for contraction on solidification, method 103 of Fed. Test Method Std. No. 101. After cooling the boxes shall be weighed to the nearest tenth of a gram and immersed in water for 24 hours at room temperature. The boxes shall be removed, wiped dry of surface water with a soft rag, and weighed.

4.5.4 *Corrosion.*

4.5.4.1 *Preparation of test panels.* Two panels shall be prepared from each of the following metals:

(a) Steel conforming to Fed. Spec. QQ-S-636.

(b) Copper conforming to Fed. Spec. QQ-C-576.

(c) Aluminum conforming to Fed. Spec. QQ-A-355.

The panels shall be 2 by 3 by ⅛ inch with rounded corners and edges. A ¼-inch hole shall be drilled in each panel ½ inch from the edge of the panel on the center line of the two-inch width. The panels shall be polished using a No. 320A wet or dry silicon carbide paper; in addition the copper and aluminum panels will be polished with No. 00 steel wool. After polishing the panels shall be washed with a volatile solvent such as Stoddard Solvent and wiped dry in such a manner as to leave no lint on the surface. Panels shall be fitted with suspension hooks made by passing a double strand of copper, monel or stainless steel wire through the hole, twisting it securely in place and forming an eyelet approximately ⅜-inch in diameter above the 0.5-inch twisted shank. The panels shall be suspended, washed with warm (120°F.) 50 percent aqueous alcohol solution, rinsed with boiling absolute alcohol and allowed to attain room temperature in a calcium chloride desiccator (70°-77°F.).

4.5.4.2 *Test procedure.* The panels prepared in 4.5.4.1 shall be lowered until completely immersed in a molten bath of the sealing compound at 210° ± 2°F. for a period of 24 hours. The panels shall then be removed, solvent rinsed and dried. The steel strips shall show no trace of corrosion. The copper and aluminum panels shall show no more staining and corrosion than similar clean, polished panels heated for a period of 24 hours in an oven at 210° ± 2°F.

4.5.5 *Setting time.* Four conditioned strips of paper shall be coated as specified in method 193 of Fed. Test Method Std. No. 101. The coating on each side of the paper shall be not less than 1/64 inch in thickness and the total thickness of the two films not less than 1/32 inch. The coated strips shall be cooled at 70° ± 5°F. for a period of one and one-half (1.5) minutes and then immediately placed in a stack with the coated surfaces fully in contact. A ½-pound bar weight 1 inch by 6 inches in base dimensions shall be placed lengthwise on the center portion of the stack of four specimens for 5 minutes. After removing the weight, it shall be possible to separate the coated strips without damaging the coating on any one of the strips. Slight adherence between uneven portions of the coating shall not be cause for rejection.

4.6 Rejection.

4.6.1 *Rejection by examination of filled containers.* Any container in the sample having one

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or more defects or under required fill shall be rejected and if the number of defective containers in any sample exceeds the acceptance number of the appropriate sampling plan of Military Standard MIL–STD–105, the lot shall be rejected.

4.6.2 *Rejection by testing.* A lot shall be rejected for failure of the sample to comply with any requirement of section 3 when tested in accordance with 4.5.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packing.

5.1.1 *Level A.* Unless otherwise specified, dipcoat sealing compound shall be packed in units of not more than three 50-pound slabs in cleated plywood, nailed wood, wirebound wood, wood-cleated fiberboard boxes or in bulk form in slack barrels, kegs or fiber drums.

5.1.1.1 Cleated plywood, nailed wood, wirebound wood and wood-cleated fiberboard boxes shall conform to the requirements of Federal Specification PPP–B–601 (overseas type), PPP–B–621 (class 2), PPP–B–585 (class 3) and PPP–B–591 (overseas type) respectively. Slabs shall be separated by means of a sheet of not less than 30-pound (24x36x500) basis weight dry waxed kraft or sulphite paper. Boxes shall be closed and strapped in conformance with the requirements of the appendix of the applicable box specification.

5.1.1.2 Slack barrels, kegs, and fiber drums shall conform to the requirements of Federal Specifications PPP–B–41 (type II), NN–K–231, and PPP–D–723, respectively. Closure and reinforcement shall be in accordance with the appendix to the applicable container specification.

5.1.2 *Level B.* Unless otherwise specified, dipcoating compound shall be packed in units of two 50-pound slabs in fiberboard boxes, or in bulk form in fiber drums, slack barrels, or kegs.

5.1.2.1 Fiberboard boxes shall conform to the requirements of Federal Specification PPP–B–636 (class 1).

5.1.2.2 Fiber drums shall conform to Federal Specification PPP–D–723 except that drums will not be required to meet the wet tests, and the dry compression load may be reduced to 750 pounds. Slack barrels and kegs shall conform to the requirements of Federal Specifications PPP–B–41 (type I) and NN–K–231, respectively. The gross weight shall not exceed approximately 200 pounds.

5.1.3 *Level C.* The dipcoat compound shall be packed in a manner which will assure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Shipping containers or method of packing shall comply with common carrier regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* Unless otherwise specified, shipments shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The sealing compound covered by this specification is intended for sealing by dipping or wrappings or packages for domestic and overseas shipment.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Level of packing required (see section 5).
(*c*) Size and weight of blocks desired.
(*d*) Special markings if required (see 5.2).

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6.3 Supersession data. This specification includes the requirements of Military Specification JAN–P–115 dated November 30, 1944.

6.4 Toxicity. Any questions raised regarding toxicity should be referred by the procuring agency to the departmental medical authority. In the case of Army procurement, The Surgeon General will act as advisor to the procuring agency.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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